

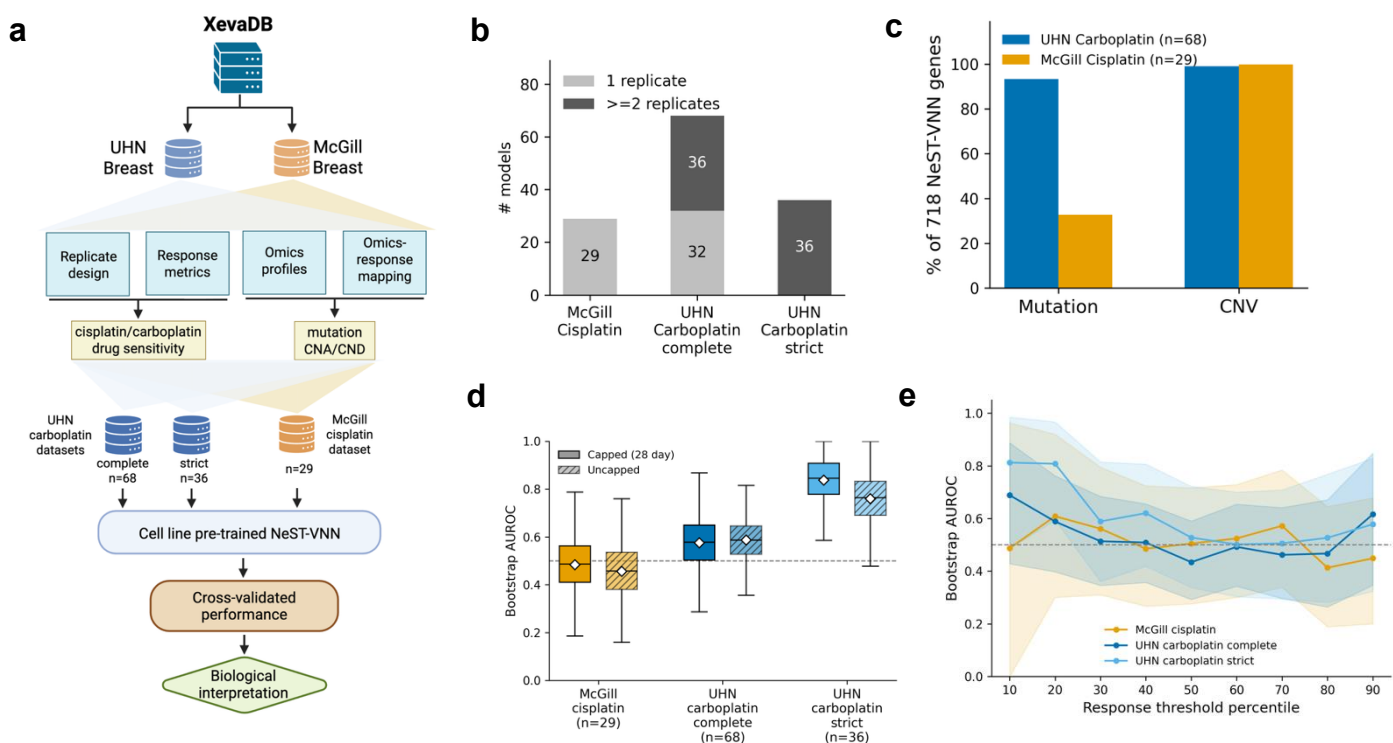


Figure 4. Cross-platform validation of NeST-VNN on independent PDX cohorts.

(a) Case study three workflow: drug-response, mutation, and copy-number data from the McGill (cisplatin, $n = 29$) and UHN (carboplatin; complete cohort $n = 68$, replicate-supported "strict" cohort $n = 36$) breast PDX cohorts are used as external input to a cell-line-pretrained NeST-VNN ensemble, validated against a growth-curve response label. (b) Number of models with exactly one versus two or more treatment-arm replicate mice, for each cohort/cohort definition. (c) Fraction of the 718-gene NeST-VNN input panel with recorded mutation and copy-number (CNV) calls in each cohort's raw molecular profiles. (d) Bootstrap 95% confidence intervals for NeST-VNN's AUROC under the capped (28 day window, primary) and uncapped (full observed duration) response-label definitions, for each of the three cohort definitions; dashed line, chance performance. (e) NeST-VNN performance across alternative response definitions. AUROC was evaluated across cohort-specific treatment-arm slope percentile cutoffs, with models below each cutoff classified as sensitive. Lines represent the McGill cisplatin, UHN carboplatin complete and UHN carboplatin strict cohorts, shaded regions indicate bootstrap 95% confidence intervals, and the dotted line indicates chance performance (AUROC = 0.5).